National Digital Medical Record System (NDMRS)

Milestone 1 — Proposal & Architectural Requirements

CSE4203: Principles of Software Architecture

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Field	Value
Project name	National Digital Medical Record System (NDMRS)
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1. System Overview

Problem. Patient health data is typically scattered across clinics, hospitals, labs and pharmacies. A doctor treating a patient often cannot see past encounters, allergies, lab results or active prescriptions, which leads to repeated tests, prescribing errors and slow emergency care. Existing lab systems are legacy and cannot be replaced overnight.

Solution. NDMRS is a national digital medical record system. It keeps one chronological record per patient (demographics, blood type, allergies, chronic conditions, vaccinations, encounters, diagnoses, clinical notes, prescriptions and lab results) and exposes it to the right people through role-specific dashboards.

Who uses it. Five clinical and patient-facing roles, each with its own dashboard: **Patient**, **Doctor**, **Lab Technician**, **Pharmacist** and **Emergency Responder**. A **System Administrator** manages accounts and roles.

Also. Nurses support encounters and record observations. NDMRS integrates with an existing legacy laboratory system and with the Ministry of Health national database.

Main capabilities. Account and identity management; patient records with a chronological timeline; clinical encounters; prescriptions and dispensing; laboratory orders and results; emergency access to critical data; notifications; and a complete audit trail.

Why it is architecturally interesting.

- *Security vs. usability.* Medical data is highly sensitive and every access must be authorized and audited, yet in an emergency a responder needs fast access to a patient's allergies and blood type (break-glass access).
- *Modifiability vs. performance.* The design is a layered system (Bootstrap, Presentation, Application Services, Domain, Infrastructure) with 8 bounded modules (Identity, Patient Records, Clinical, Lab, Pharmacy, Emergency, Notification, Audit). Loose coupling through an in-memory domain-event bus makes the system easy to extend, but adds indirection that must not hurt response time.
- *Integration.* A legacy lab system must be wrapped (Adapter) and notifications must be deliverable over several channels (Strategy).
- *Rich design-pattern use.* The architecture applies 14 patterns, including the 5 required ones: Singleton (ApplicationContext), Abstract Factory (role dashboards), Factory Method (repositories), Iterator (patient timeline) and Observer (event bus).
- *Many roles over one record.* The eight bounded modules share one domain model, and each role works on it through its own dashboard.
- *Lifecycle workflows.* Encounters, prescriptions and lab orders move through states, and several parties must react to each change.

Initial architectural direction. The team's design work starts from a layered architecture with bounded modules (Bootstrap, Presentation, Application Services, Domain, Infrastructure); a Java Swing GUI using Model-View-Presenter; repository-based, swappable file persistence; and domain events published over an in-memory event bus. This is a starting hypothesis; the final choice will be justified against the quality attributes in Milestone 2.

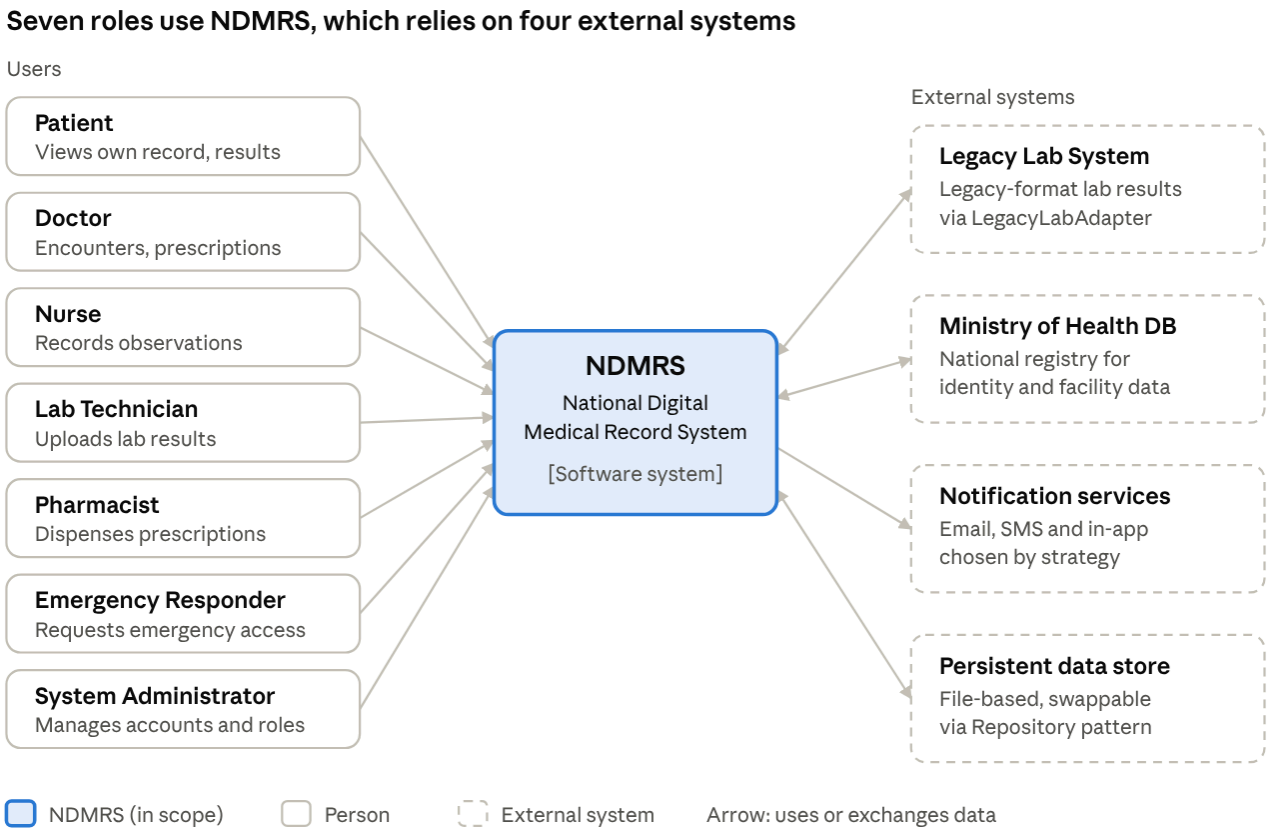
2. Stakeholders and Concerns

#	Stakeholder	Key concerns
1	Patients	Privacy of their data; easy read-only access to their own history, prescriptions and results; being notified of new results/prescriptions
2	Doctors	Complete and fast view of a patient's history; safe prescribing (allergy awareness); simple encounter workflow
3	Lab technicians	Easy upload of lab results; compatibility with the existing legacy lab equipment/system; accurate order tracking
4	Pharmacists	Verified prescriptions; preventing double dispensing; clear prescription state (pending, dispensed, expired, cancelled)
5	Emergency responders	Immediate access to critical data (blood type, allergies, conditions) under time pressure, with that access fully logged
6	System administrators / operators	Account and role management; deployment simplicity; monitoring; backup of the data store
7	Health ministry / regulators	Auditability of every access and change; compliance with health-data privacy rules; data integrity
8	Developers / maintainers (the team)	Modifiability, testability, clear module boundaries, ability to swap storage, notification and lab integrations

#	Stakeholder	Key concerns
9	Healthcare facilities (hospitals, labs, pharmacies)	Interoperability with existing lab systems; reliability; minimal disruption to current workflows
10	Nurses	Simple screens; correct patient identification; limited but sufficient permissions
11	Ministry of Health (business owner)	National interoperability; cost; accountability and reporting

3. System Context (C4 Level 1)

NDMRS is shown as a single system. The diagram below is the C4 Level 1 view; its source file will be kept in the repository.



C4 Level 1 context · 7 users, 4 external systems

Seven roles use NDMRS; it exchanges data with the legacy lab, the Ministry of Health database and the data store, and sends messages through the notification services.

Users (people)

Actor	Interaction with NDMRS
Patient	Logs in, views own record, timeline, prescriptions, lab results; receives notifications
Doctor	Opens/closes encounters, writes diagnoses, notes and prescriptions, orders labs, searches patients
Lab Technician	Views lab orders and uploads lab results
Pharmacist	Views prescriptions and dispenses them
Emergency Responder	Requests emergency access to a patient's critical information
System Administrator	Creates and manages accounts, roles and permissions
Nurse	Supports encounters and records observations

External systems

External system	Relationship
Legacy Lab System	Provides/accepts lab results in a legacy format; NDMRS wraps it through the LegacyLabAdapter
Notification services (Email, SMS, In-app)	NDMRS sends messages to patients when results are ready, prescriptions are written or dispensed, etc.
Persistent data store (file-based, swappable via the Repository pattern)	Stores accounts, patients, encounters, prescriptions, lab orders and the audit log
Ministry of Health database	National registry used to validate identity and exchange patient and facility data

4. Functional Requirements

Priorities use MoSCoW. The last column marks what the Milestone 3 prototype will implement.

ID	User story	Priority	In M3 prototype
FR1	As any user, I can log in with a role-based account so that I only see what my role allows.	Must	Yes
FR2	As a doctor, I can open an encounter for a patient and close it when finished.	Must	Yes
FR3	As a doctor, I can view a patient's full chronological timeline (encounters, diagnoses, notes, prescriptions, lab results).	Must	Yes
FR4	As a doctor, I can write a prescription within an encounter.	Must	Yes
FR5	As a pharmacist, I can dispense a valid prescription and the system prevents dispensing it twice.	Must	Yes
FR6	As a lab technician, I can upload a lab result for a lab order.	Must	Yes
FR7	As an emergency responder, I can get emergency access to a patient's critical data, and the access is recorded.	Must	Yes
FR8	As an auditor/admin, I can see an audit log of who accessed or changed what.	Must	Yes
FR9	As a patient, I can view my own record, prescriptions and lab results (read-only).	Should	Yes
FR10	As a patient, I get a notification when a lab result or prescription is added.	Should	Yes (one channel)
FR11	As a doctor, I can search for patients with combined criteria (e.g., age over 60 AND diabetic AND NOT allergic to penicillin).	Should	Yes
FR12	As a doctor, I can order a lab test and track its state (ordered, in progress, completed, verified).	Should	Partially

ID	User story	Priority	In M3 prototype
FR13	As an administrator, I can create, deactivate and assign roles to accounts.	Should	No
FR14	As a lab technician, results from the legacy lab system are imported automatically.	Could	Yes (simulated legacy lab)
FR15	As a patient, I can receive notifications by email or SMS in addition to in-app.	Could	No
FR16	As a user, I can use the system in Arabic and English.	Won't (this release)	No
FR17	As a clerk or doctor, I can register a new patient record.	Must	Yes
FR18	As a nurse, I can record observations and update the encounter state.	Could	No
FR19	As the system, I can synchronize with the Ministry of Health database.	Could	No

5. Quality Attribute Scenarios

Ten scenarios covering security, usability, performance, modifiability, availability, interoperability and testability.

QAS-1 — Security (access control)

Part	Description
Source	A logged-in pharmacist (or an attacker with a valid pharmacist account)
Stimulus	Attempts to read a patient's clinical notes or write a prescription
Artifact	Application services layer (command handlers)
Environment	Normal operation
Response	The request is rejected by the authorization decorator before reaching the domain; the attempt is written to the audit log

Part	Description
Response measure	100% of unauthorized requests are rejected in a test suite of at least 50 role/permission combinations; each rejection is logged within 1 second

QAS-2 — Security / Usability trade-off (emergency access)

Part	Description
Source	Emergency responder
Stimulus	Requests emergency access to an unconscious patient's record
Artifact	Emergency module and authorization policy
Environment	Emergency, normal load
Response	The system grants read-only access to critical data only (blood type, allergies, chronic conditions), publishes an EmergencyAccessGranted event, and notifies the patient
Response measure	Access is granted in at most 3 user actions and under 2 seconds; 100% of such accesses appear in the audit log with responder ID and timestamp

QAS-3 — Performance

Part	Description
Source	Doctor
Stimulus	Opens a patient's timeline (merged chronological view of encounters, prescriptions and lab results)
Artifact	PatientTimeline iterator, repositories and presentation layer
Environment	Normal operation, 50 concurrent users, patient with 500 historical events
Response	The timeline is displayed

Part	Description
Response measure	95% of requests complete in under 1 second; the first page of events appears in under 500 ms

QAS-4 — Modifiability

Part	Description
Source	Developer
Stimulus	Needs to add a new notification channel (e.g., WhatsApp) or replace the file-based storage with a database
Artifact	Notification module (Strategy) and persistence layer (Repository, Factory Method)
Environment	Design time / maintenance
Response	The change is made by adding a new class that implements the existing interface and changing configuration only; no domain or presentation code changes
Response measure	Completed by one developer in at most 2 person-days; at most 2 existing source files modified; all existing tests still pass

QAS-5 — Availability / Reliability

Part	Description
Source	Legacy lab system
Stimulus	Becomes unreachable or returns malformed data while a technician uploads or imports a result
Artifact	LegacyLabAdapter and lab integration module
Environment	Normal operation
Response	The adapter fails safely: the error is reported to the technician, the result is queued for retry, and the rest of the system (records, prescriptions) continues working

Part	Description
Response measure	No effect on other use cases (0 failed requests elsewhere); failed imports are retried automatically for up to 3 attempts; the system remains 99% available during a 1-hour simulated lab outage

QAS-6 — Testability / Auditability

Part	Description
Source	Tester or regulator
Stimulus	Needs to verify that every important change was recorded
Artifact	Event bus, audit module and command handlers
Environment	Test environment with in-memory repositories
Response	Each command (write prescription, dispense, upload result, close encounter, grant emergency access) publishes a domain event that the audit handler records
Response measure	100% of the 5 domain-event types produce an audit entry; domain logic can be unit-tested without the GUI, with at least 70% line coverage on the domain and application layers

QAS-7 — Modifiability (new user role)

Part	Description
Source	Developer
Stimulus	Adds a new user role (for example Radiologist) with its own dashboard
Artifact	Presentation layer and dashboard factory
Environment	Design time
Response	The new role is added without changing existing role code

Part	Description
Response measure	At most 3 existing files modified; completed in under 1 working day; all existing tests still pass

QAS-8 — Availability (notification gateway outage)

Part	Description
Source	Notification gateway
Stimulus	Becomes unreachable while a doctor writes a prescription
Artifact	Notification module and event bus
Environment	Normal operation, gateway down
Response	The prescription is saved; the notification is queued and retried
Response measure	100% of prescriptions are saved during a 30-minute gateway outage; queued alerts are delivered within 5 minutes after recovery

QAS-9 — Interoperability (legacy lab results)

Part	Description
Source	Legacy lab system
Stimulus	Returns results in its own format
Artifact	Lab integration adapter
Environment	Normal operation
Response	Results are converted to the domain result model and attached to the correct order
Response measure	99% of 100 sample results are mapped correctly; replacing the adapter requires no change in the Domain layer

QAS-10 — Security (auditability of every access)

Part	Description
Source	Any authenticated user
Stimulus	Reads or modifies a patient record
Artifact	Audit module and record access path
Environment	Normal operation
Response	An audit entry is written with user, role, patient, action and timestamp
Response measure	100% of 1,000 scripted accesses have a matching audit entry; entries cannot be edited through the application

6. Constraints

Technical

- Implemented in Java; the GUI uses Java Swing with the Model–View–Presenter pattern.
- Persistence is file-based in the prototype, but must be accessed only through repository interfaces so it can be swapped.
- Components communicate through an in-memory event bus (Observer); no external message broker in the prototype.
- The legacy lab system cannot be modified; it must be wrapped by an adapter.
- The architecture must use the five required patterns (Singleton, Abstract Factory, Factory Method, Iterator, Observer).
- Passwords must be stored only as hashes, using a replaceable hashing strategy.
- The prototype must run on a standard student laptop with no paid cloud services.

Business / regulatory

- Patient data is confidential; every access and change must be auditable.
- No real patient data is used; only synthetic data in the prototype.
- Only free and open-source libraries and tools may be used.

Organizational

- Team of 4 students.

- Milestones: M1 due October 23, 2026, then M2 (architecture document), M3 (prototype), M4 (evaluation and presentation).
- All work is submitted through GitHub as PDFs named TeamName_Project'sTitle.pdf, with diagram source files in the repository.
- Late submissions lose 10% per day, up to 3 days.

7. Architecturally Significant Requirements (ASRs)

Prioritized table

Business value and architectural impact are rated High /Medium /Low .

Rank	Requirement	Business value	Architectural impact	Source
1	Role-based authorization and audit of all access (QAS-1, FR1, FR8)	H	H	Security
2	Emergency break-glass access with logging (QAS-2, FR7)	H	H	Security vs. usability
3	Swappable storage, notification and lab integrations (QAS-4, FR14, FR15)	M	H	Modifiability
4	Timeline performance for large records (QAS-3, FR3)	H	M	Performance
5	Fault isolation from the legacy lab system (QAS-5, FR14)	M	H	Availability
6	Testability without the GUI (QAS-6)	M	M	Testability
7	Patient notifications (FR10)	M	L	Functional
8	Bilingual UI (FR16)	L	L	Usability

Justification of the top 5

1. **Authorization and audit (security).** This affects every command and every query, so it must be built into the structure and not added later. It drives the layered style and the Decorator pattern: Authorization, Audit and Logging decorators wrap every command handler, so the Domain layer stays free of security code.
2. **Emergency access (security vs. usability).** This is the main competing-quality-attribute pair of the project. It requires a dedicated Emergency module, a restricted read-only view and mandatory audit, and it shapes the Identity and Access design (roles and permissions).
3. **Modifiability.** Storage, notification channels and the lab system are all expected to change. This drives the use of bounded modules, the Repository pattern with Factory Method, Strategy for notifications and hashing, and the Adapter for the legacy lab. The Observer-based domain-event bus lets new reactions (audit, notifications) be added without changing the publisher.
4. **Performance of the timeline.** Merging encounters, prescriptions and lab results chronologically must stay fast as records grow. This drives the Iterator-based Patient Timeline and the need for efficient repository queries (Specification pattern for filtered searches).
5. **Availability through fault isolation.** A failing legacy lab must not stop clinical work. This drives putting the lab behind an Adapter and a gateway interface, and handling failures and retries at the module boundary.

Utility tree (summary)

Quality attribute	Refinement	Scenario	(Value, Impact)
Security	Access control	QAS-1	(H, H)
Security	Emergency access	QAS-2	(H, H)
Modifiability	Change integrations	QAS-4	(M, H)
Performance	Timeline response	QAS-3	(H, M)
Availability	Legacy lab failure	QAS-5	(M, H)
Testability	Domain tests without GUI	QAS-6	(M, M)
Security	Complete audit trail	QAS-10	(H, H)
Modifiability	Add roles easily	QAS-7	(H, H)
Availability	Isolate notification gateway	QAS-8	(H, M)

Quality attribute	Refinement	Scenario	(Value, Impact)
Interoperability	Wrap legacy lab system	QAS-9	(M, H)

Key trade-offs to resolve in Milestone 2

1. **Security vs. usability.** Emergency access must be fast but still controlled and audited.
2. **Modifiability vs. performance.** Event-based decoupling adds indirection that must still meet the QAS-3 response time.
3. **Availability vs. consistency.** Queued notifications are delivered eventually, so alerts may lag behind saved data.

Contribution Table

Member	Contribution
Tasneem Ahmed	[Sections 1–2]
Farha Ahmed	[Section 3 and context diagram]
Mahmoud Samer	[Sections 4 and 6]
Malek Fayez	[Sections 5 and 7, references]

References

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